Supplementary Step 10 Visualization Report

Overview

Metric	Value
Included participant-sessions	36
Included questions	30
Included trials	1075
Primary robust SE type	Two-way cluster (participant + question)
Primary FE p-value	0.209749
Participant-clustered p-value	0.151850
Question-clustered p-value	0.228028
HC3 p-value	0.173034
Fixed-window benchmark p-value	0.219180
No-override p-value	0.367415
Back ROI benchmark p-value	0.224774
Leave-one-question-out p range	0.079410 to 0.412718
Leave-one-participant-out p range	0.067180 to 0.292688

Interpretation note. In the final participant-fixed-effects robustness analysis, the abstract-versus-concrete effect in the left frontal ROI was not statistically significant (beta=3.016e-08, SE=2.351e-08, p=0.209749, 95% CI [-1.793e-08, 7.824e-08]). Together with the non-significant sensitivity models, this suggests that the frontal condition tendency remains descriptive but does not receive robust statistical support in the current dataset.

Influence note. The leave-one-question-out p-values ranged from 0.079410 to 0.412718, and the leave-one-participant-out p-values ranged from 0.067180 to 0.292688. Neither influence analysis produced any p-value below 0.05.

Core Step 10 Figures

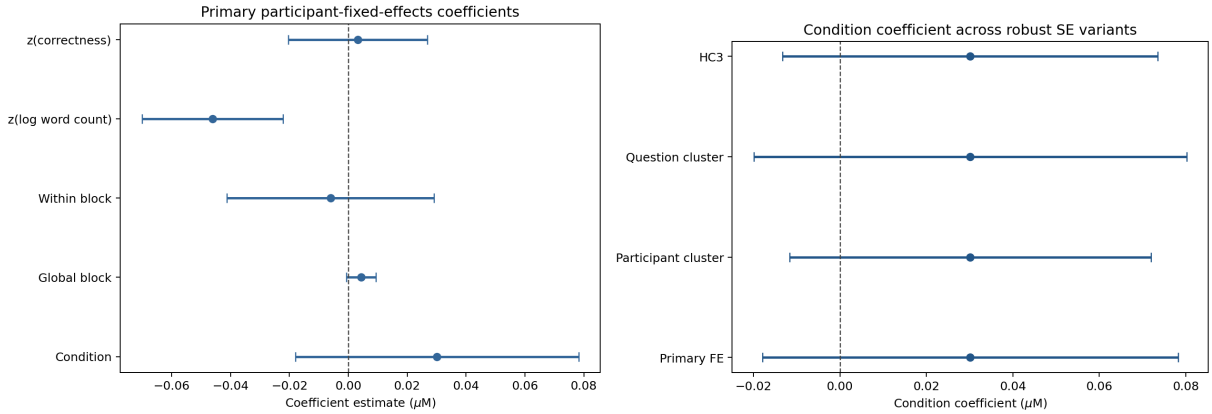

Figure 1: Primary participant-fixed-effects coefficient summary and the robust-standard-error comparison.

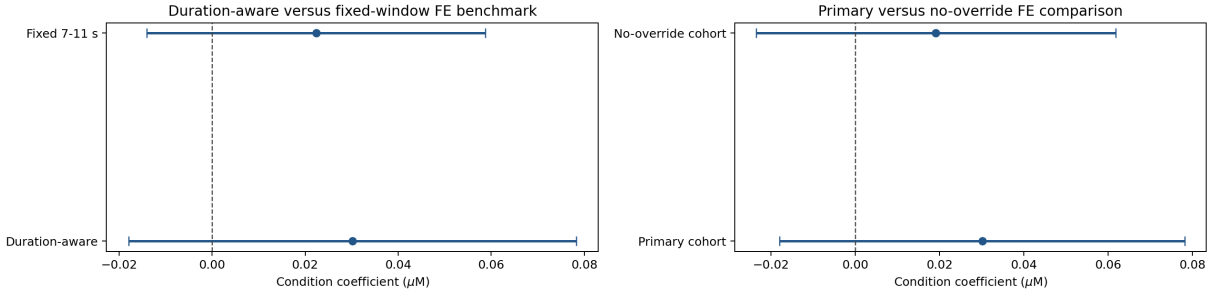

Figure 2: Duration-aware versus fixed-window benchmark and the no-override sensitivity comparison.

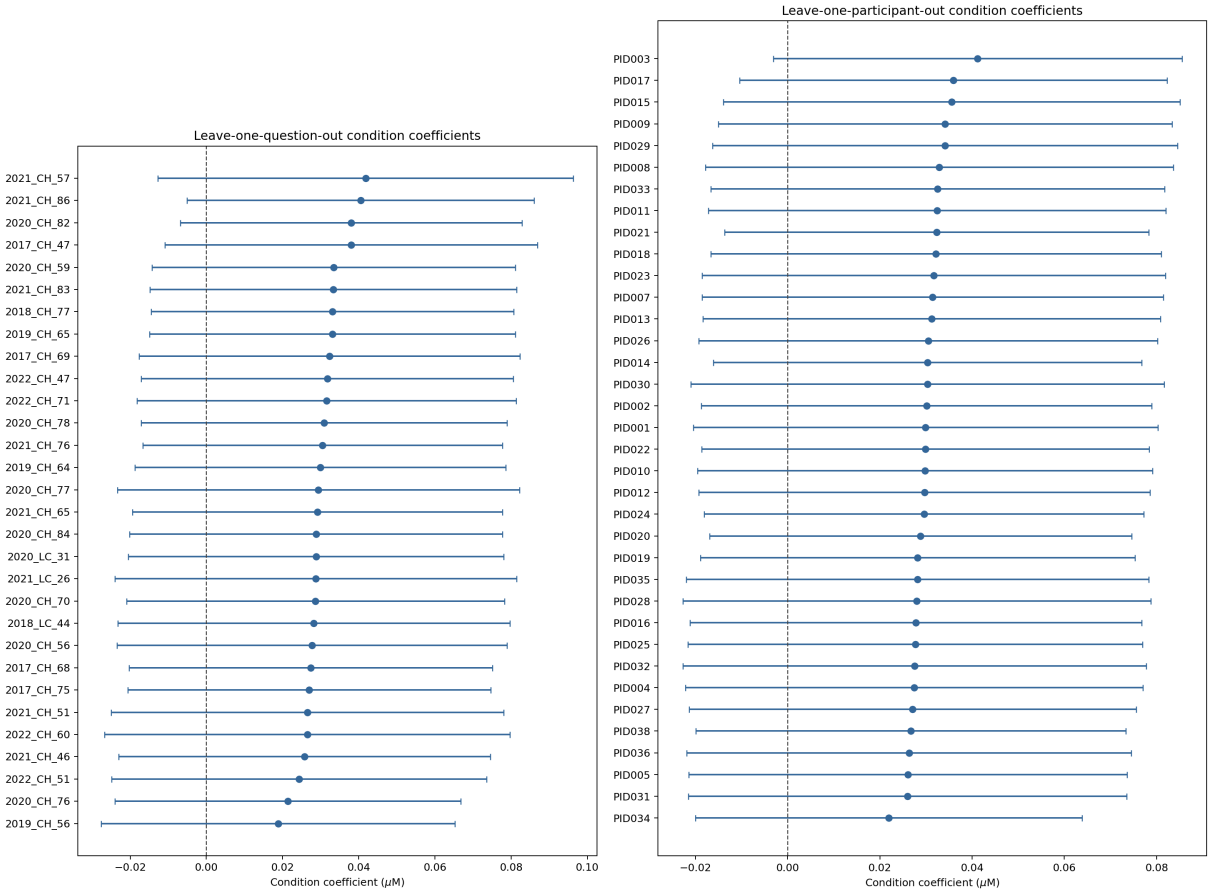

Figure 3: Condition coefficients from the leave-one-question-out and leave-one-participant-out influence analyses.

Additional Influence Diagnostics

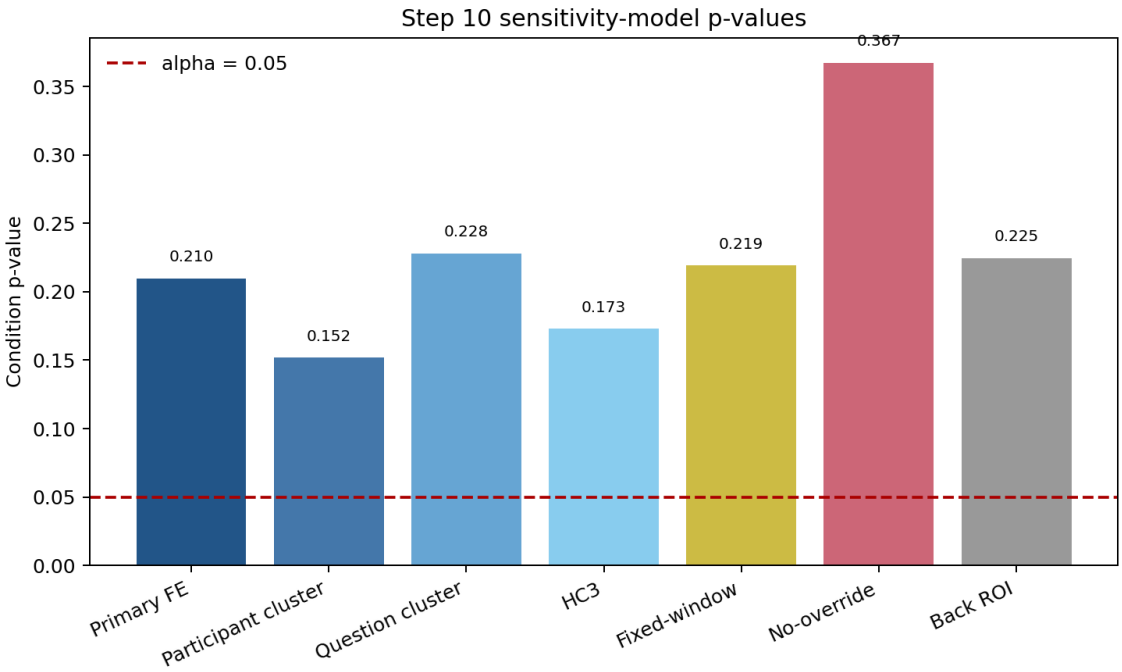

Figure 4: Condition p-values across the primary and sensitivity Step 10 models.

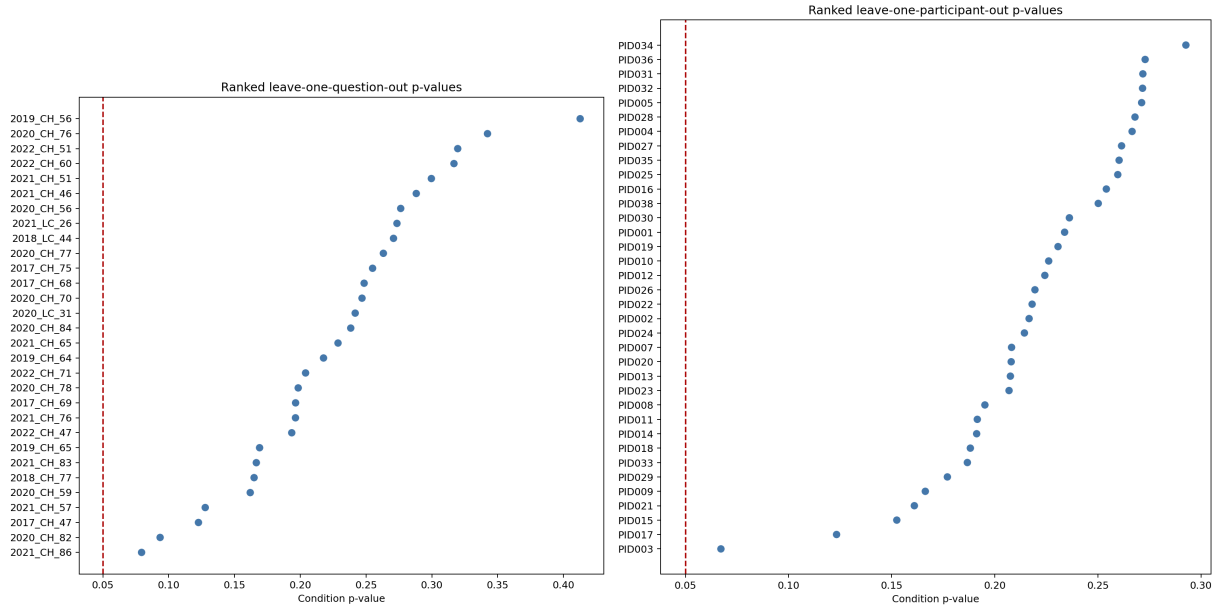

Figure 5: Ranked p-values from the leave-one-question-out and leave-one-participant-out analyses. The red line marks alpha=0.05.